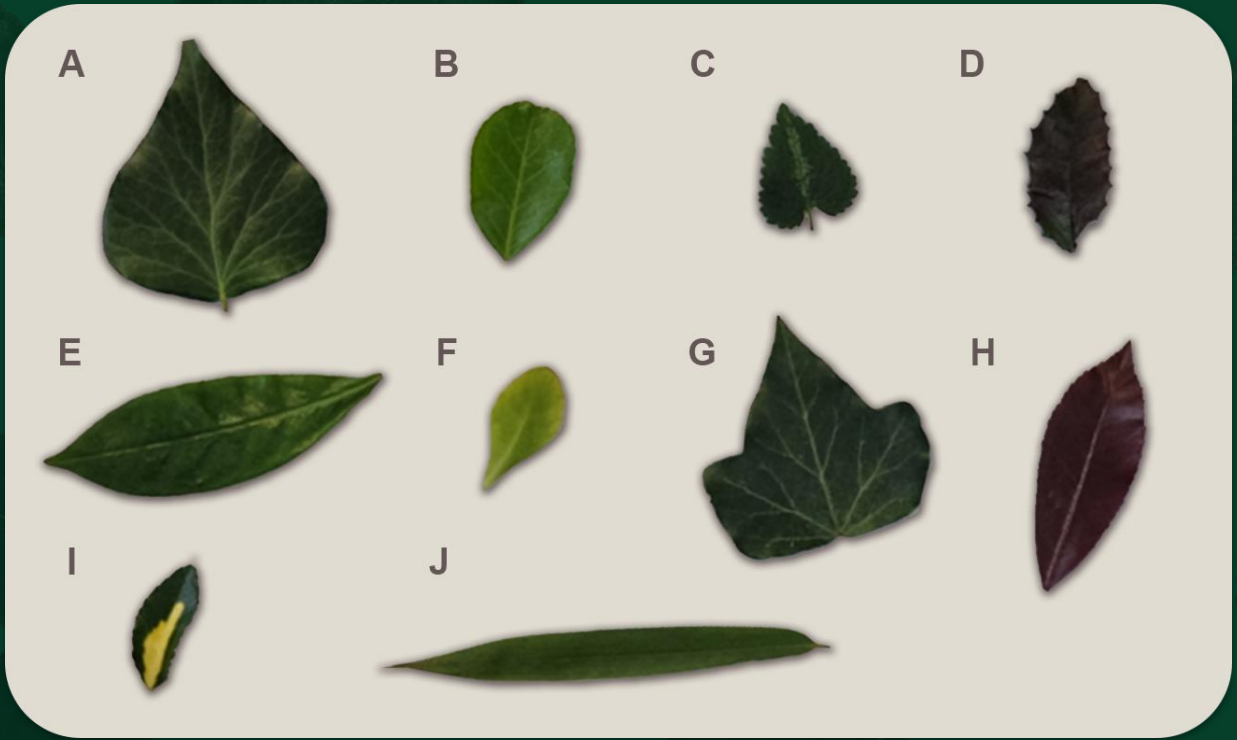
NoLeafLeftBehind

Goals

- Creating ground truth files
- Accurate segmentation
- Separate the unknowns from the leaves
- Search for optimal features to classify the leaves

Our Dataset



A	Edera del caucaso
B	Celastro
C	Falsa ortica
D	Agrifoglio
E	Oleandro
F	Telefio
G	Edera helix
H	Photinia
I	Mandrino
J	Bambù

Classi \ Foto	1	2	3	4	5	6	1	2	3	4	5	6	7	8	9	10	11	12	13
Edera del caucaso	1	0	1	2	0	1	1	0	1	1	1	2	1	1	0	1	0	1	0
Celastro	1	0	2	1	0	1	0	0	2	0	0	1	0	2	1	2	1	0	1
Falsa ortica	1	2	1	0	1	0	0	0	2	0	0	2	1	1	1	2	0	1	0
Agrifoglio	2	1	0	1	1	0	1	0	2	2	2	1	0	0	0	0	1	0	1
Oleandro	1	1	1	1	0	1	1	1	1	0	0	0	2	1	2	1	0	1	0
Telefio	1	1	0	1	1	0	1	2	0	1	0	0	1	0	1	0	2	0	2
Edera helix	0	0	2	1	1	1	1	2	0	1	1	0	1	1	0	1	0	2	0
Photinia	0	2	1	1	0	1	1	0	1	0	1	1	0	0	1	1	2	0	2
Mandrino	0	1	1	2	1	0	0	2	0	1	2	0	2	1	0	1	0	1	0
Bambù	1	1	1	0	1	1	1	1	0	1	1	2	0	0	2	0	1	1	0
Unknown	1	1	1	1	0	1	1	1	1	0	1	1	1	1	1	1	0	0	1

Table representing the class frequencies in the dataset for each image

TRAIN SET: 100 leaves + 10 unknown, ~67%

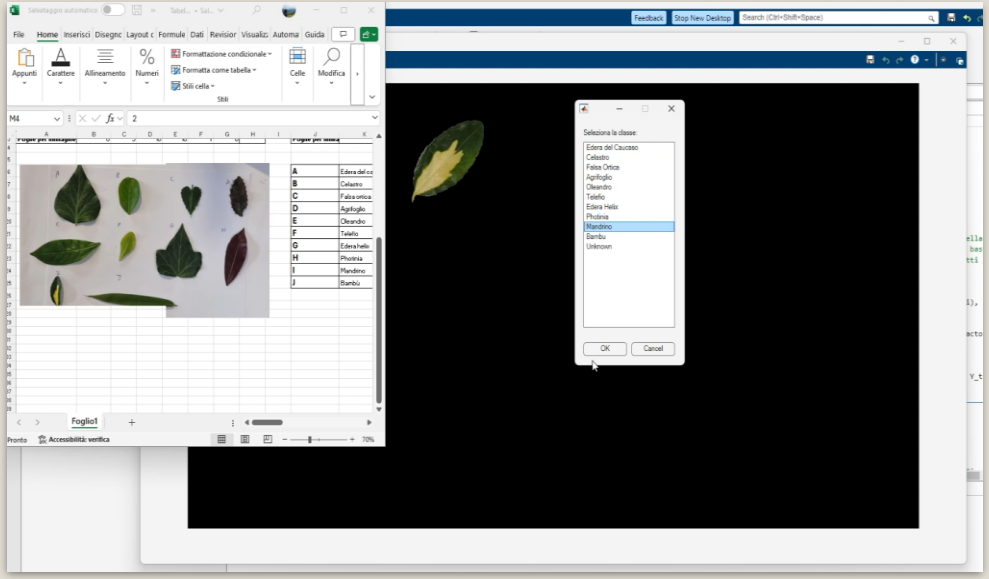
TEST SET: 50 leaves + 5 unknown, ~33%

165 objects

Assumptions

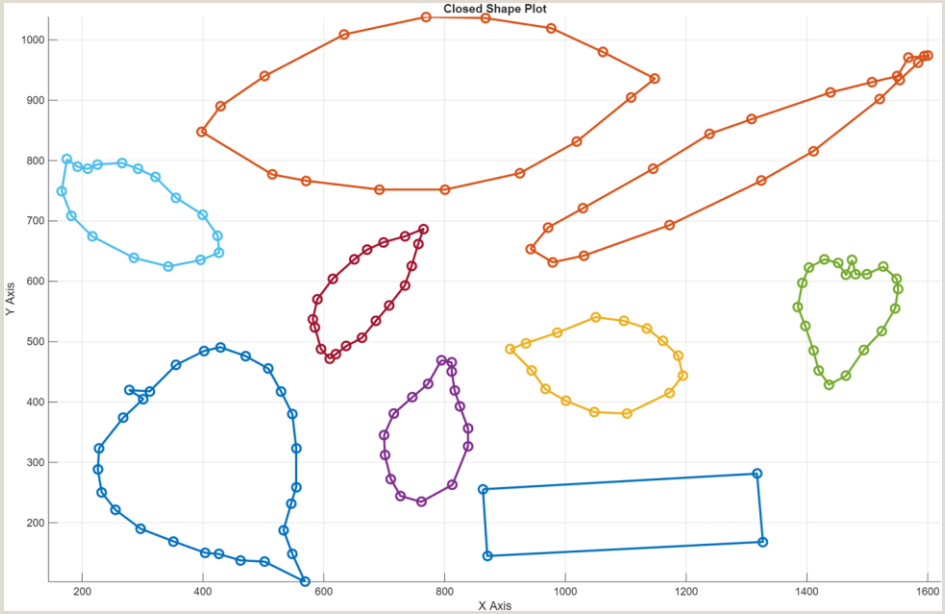
- Uniform background
- Soft shadows and noises may be present
- Pictures taken on fixed scale
- No overlapping objects

Creating Ground Truth



Manual object labelization with dialogue window for each RoI extracted from binary mask

“Unknown”



Creating manual polygon area for each objects in images in labelstud.io as metric for IoU

Pipeline

Pre-processing

Segmentation

Separation of Leaves from Unknown Objects

Leaves classification

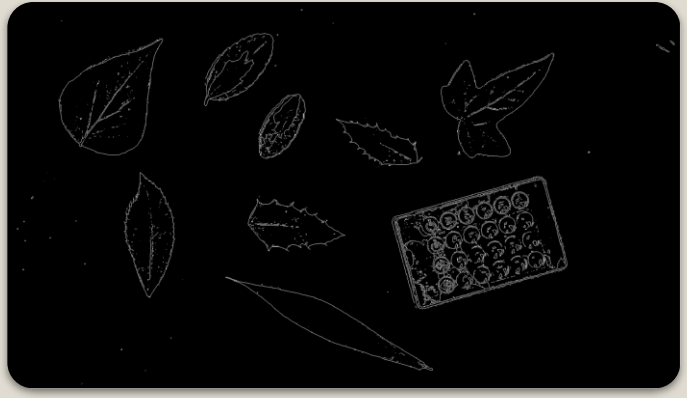
Pre-processing

Gray Scale Image

Saturation channel

Saturation channel is useful for an initial detailed border mask
Doesn’t work in ALL light conditions

Sharpened (Radius=2,Amount=0.8), gray scale



Sobel edge

We use sharpening to enhance definitions and reveal more edges, but it also amplifies noise.

Segmentation

Segmentation Pipeline

1. Mask for Details in Outer Edges
2. Mask for Areas with Excessive Details

K-Means on Gray Scale

Grayscale Image

K-means with 2 clusters

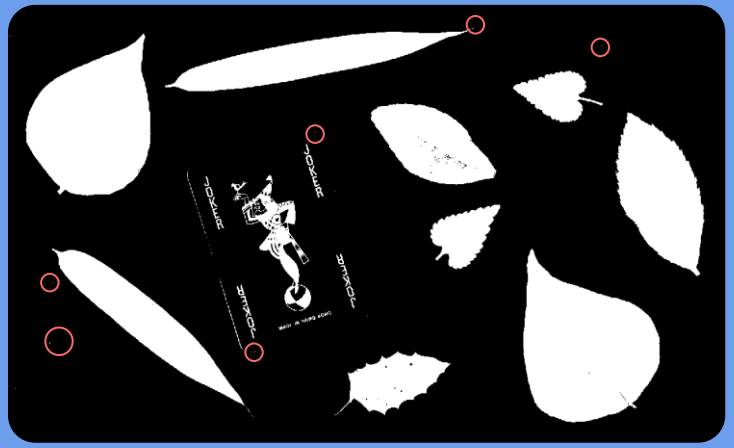
Binarize Saturation Channel

Saturation channel from HSV

Binarized mask with OTSU

Combining the

OR



Binarized images leaves some noises and holes

Filling small holes and removing noises

medfilt2(unionMasks)

Median filter for noise reduction

imfill(medianMask)

Filling leaves holes

imerode(filledMask, strel('disk', 5))

Removing bigger noises area

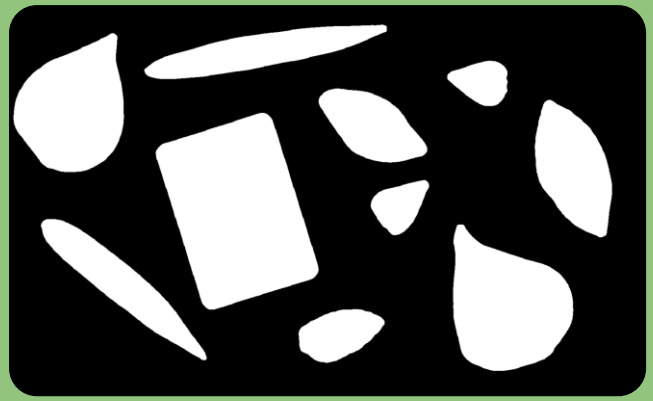
(if present)

Extract detailed objects mask with Edge

Sharpened image

Sobel edges

Dilatation and Filling with
STREL(DISK, 6)

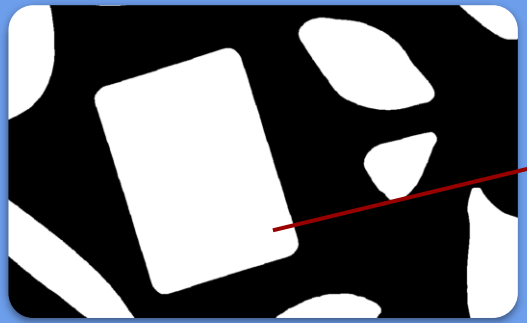


Applying median filter
and opening with STREL(DISK, 6)

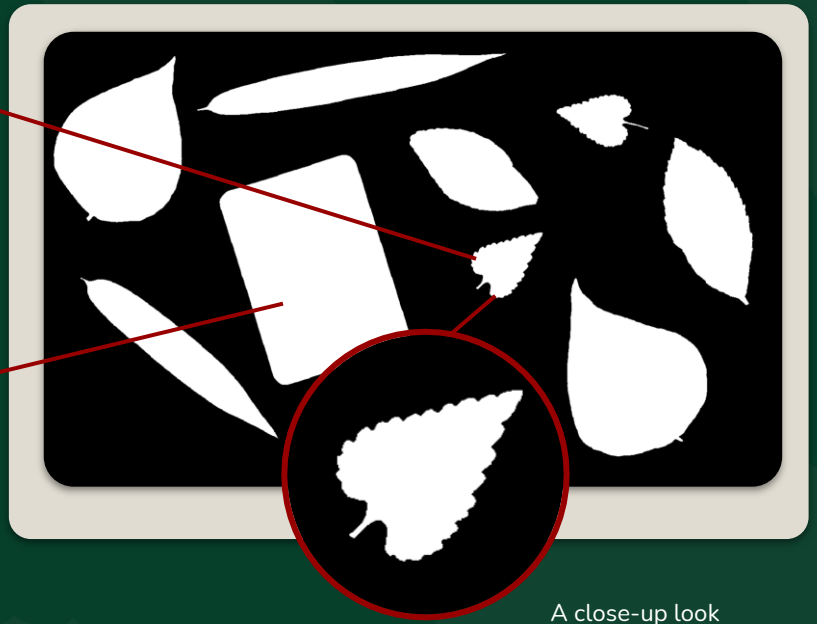
Combine the 2 masks

if **object area** / **object area** < 0.8, then UNION

Detailed mask from KMeans and Binariziation union



Fuller mask from SOBEL



A close-up look of the contour of this leaf

k mask

edge mask

Unknown Detection

Features

Gray Variance: Differentiates leaves from smooth unknowns

Color Based Feature

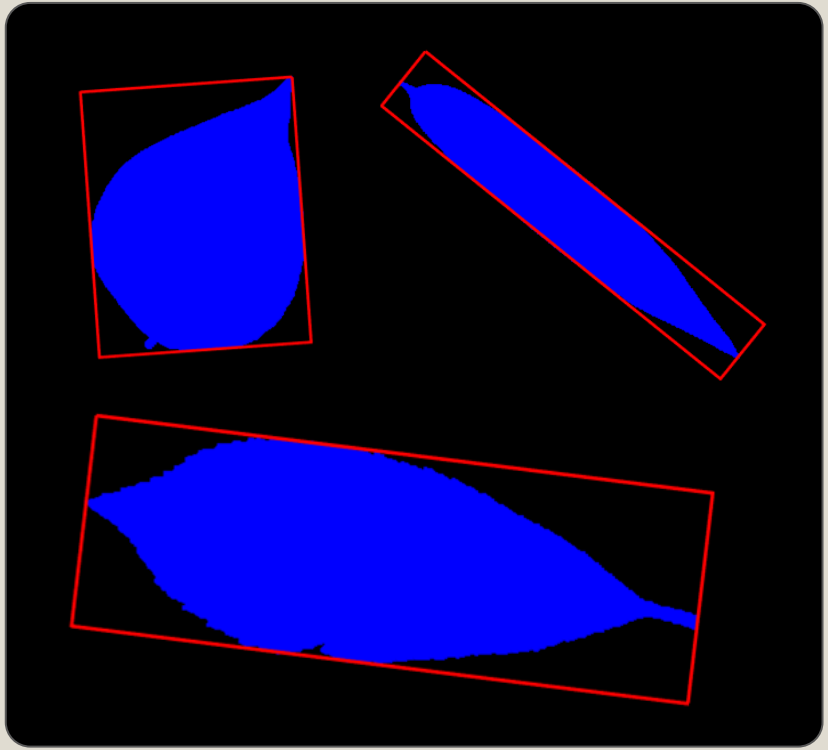
Eccentricity: Separates elongated leaves from round objects

Circularity: Helps distinguish round objects from leaves

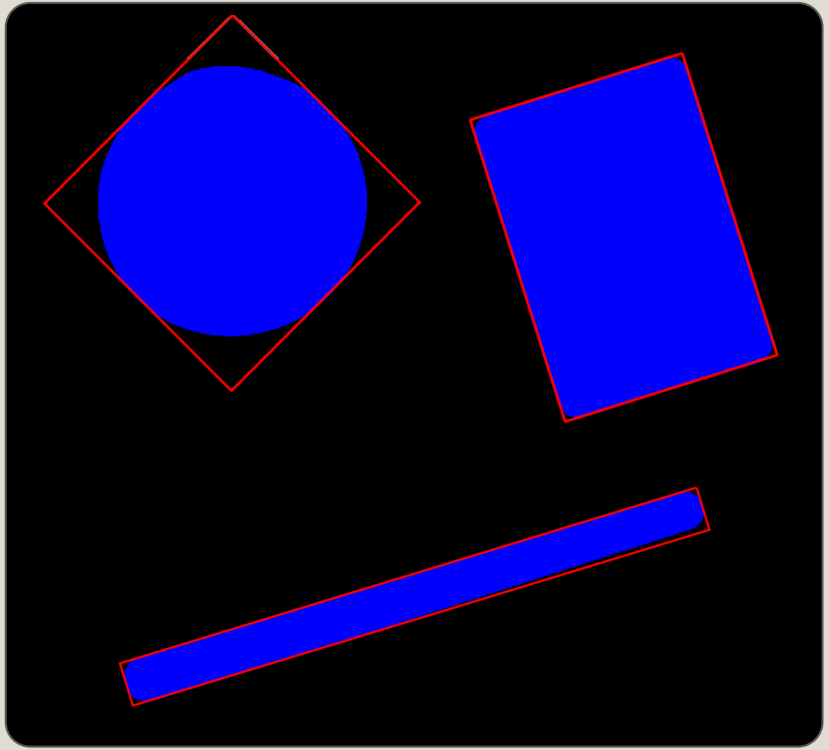
MBB ratio: % of the object's area relative to the bounding box area

Form Based Features

Calculate MMBOX/object area ratio



Non-Geometric objects have higher ratio



Geometric objects have low ratio

Leaves Classification

Features

Perimeter

Total length of an object's boundary

Eccentricity

Separates elongated leaves from round objects

Circularity

Helps distinguish round objects from leaves

First Hu Moment

Overall shape and is invariant to scale, rotation and translation

Second Hu Moment

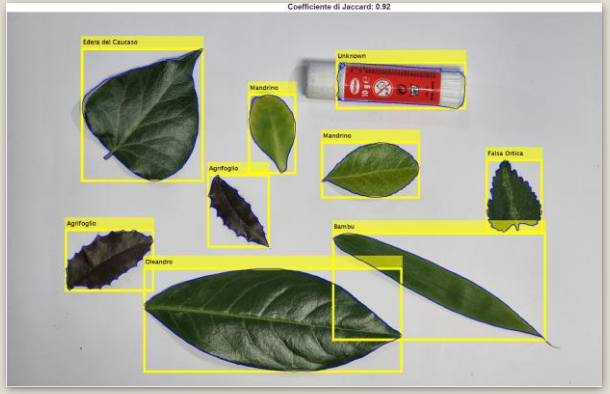
Describes symmetry and elongation of an object.

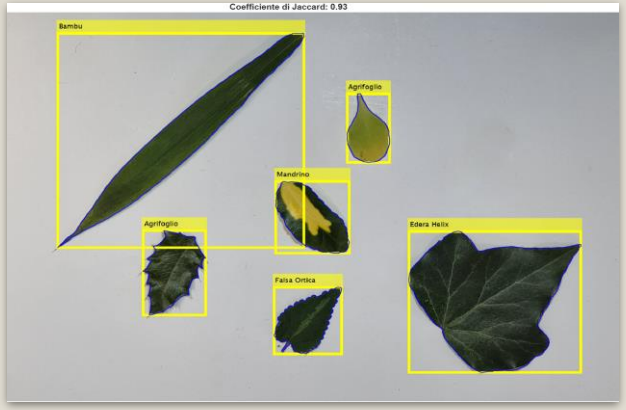
Solidity

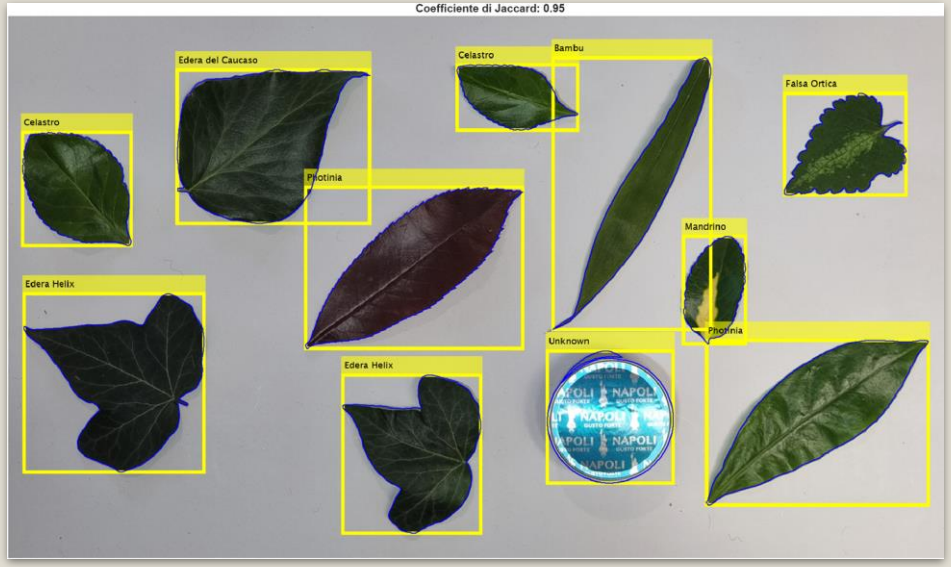
Ratio of an object's area to its convex hull area

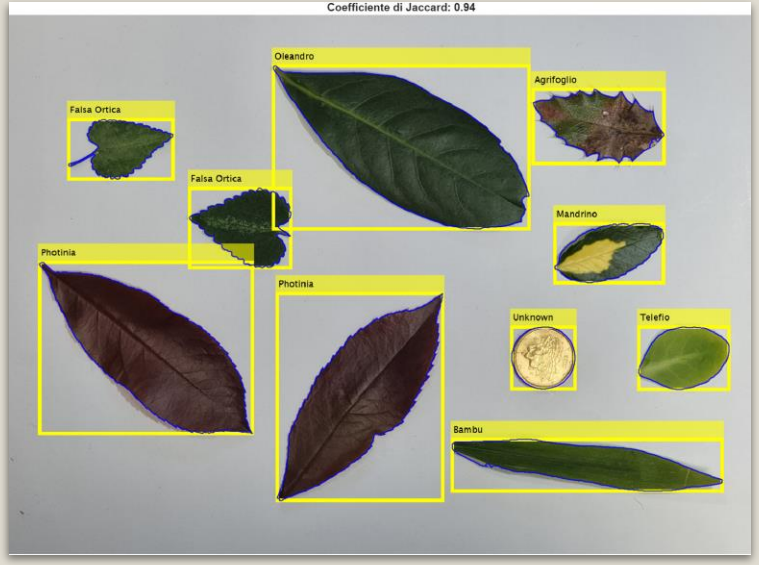
Unknown

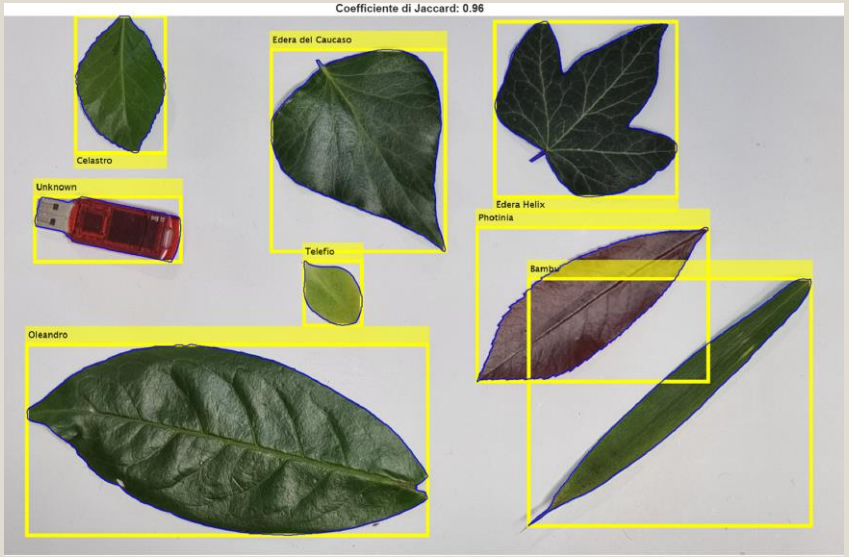
Results

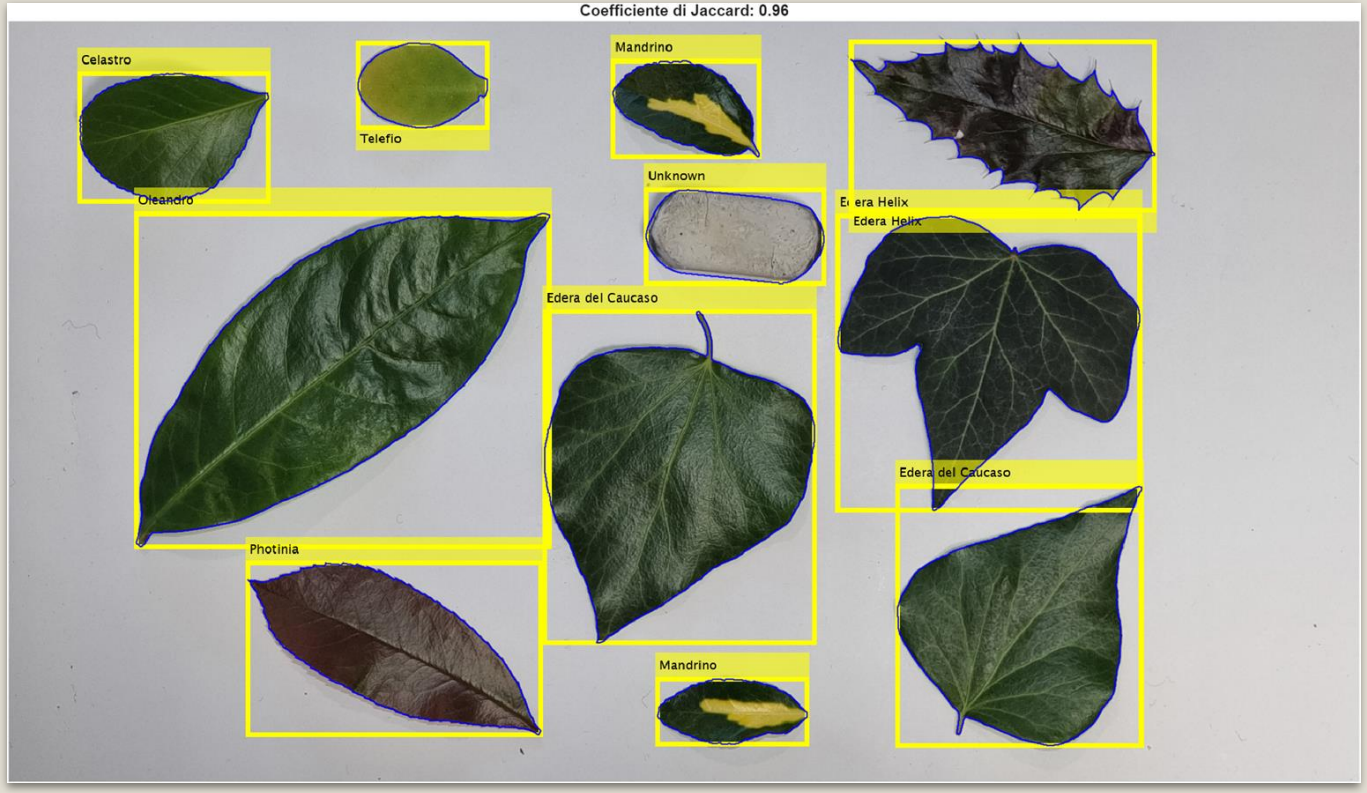

Coefficiente di Jaccard: 0.92
Edera del Caucaso
Unknown
Mandrino
Mandrino
Falsa Ortica
Agrifoglio
Agrifoglio
Bambu
Oleandro


Coefficiente di Jaccard: 0.93
Bambu
Agrifoglio
Mandrino
Agrifoglio
Edera Helix
Falsa Ortica


Coefficiente di Jaccard: 0.95
Edera del Caucaso
Celastro
Bambu
Falsa Ortica
Celastro
Photinia
Mandrino
Edera Helix
Unknown
Photinia
Edera Helix


Coefficiente di Jaccard: 0.94
Oleandro
Agrifoglio
Falsa Ortica
Falsa Ortica
Mandrino
Photinia
Photinia
Unknown
Telefio
Bambu

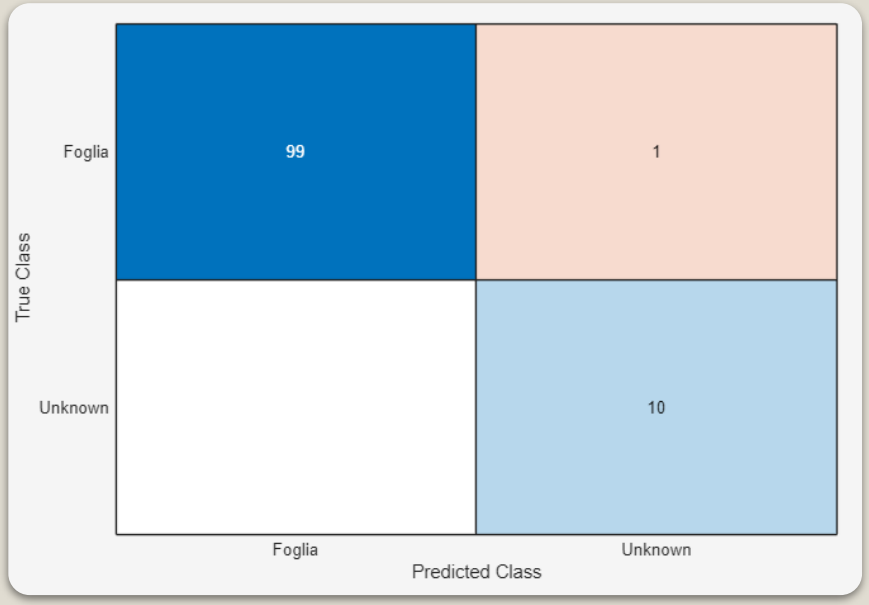

Coefficiente di Jaccard: 0.96
Edera del Caucaso
Celastro
Unknown
Edera Helix
Photinia
Telefio
Oleandro


Coefficiente di Jaccard: 0.96
Celastro
Telefio
Mandrino
Unknown
Oleandro
Edera Helix
Edera Helix
Edera del Caucaso
Edera del Caucaso
Photinia
Mandrino

CFM Model Unknown/Leaf

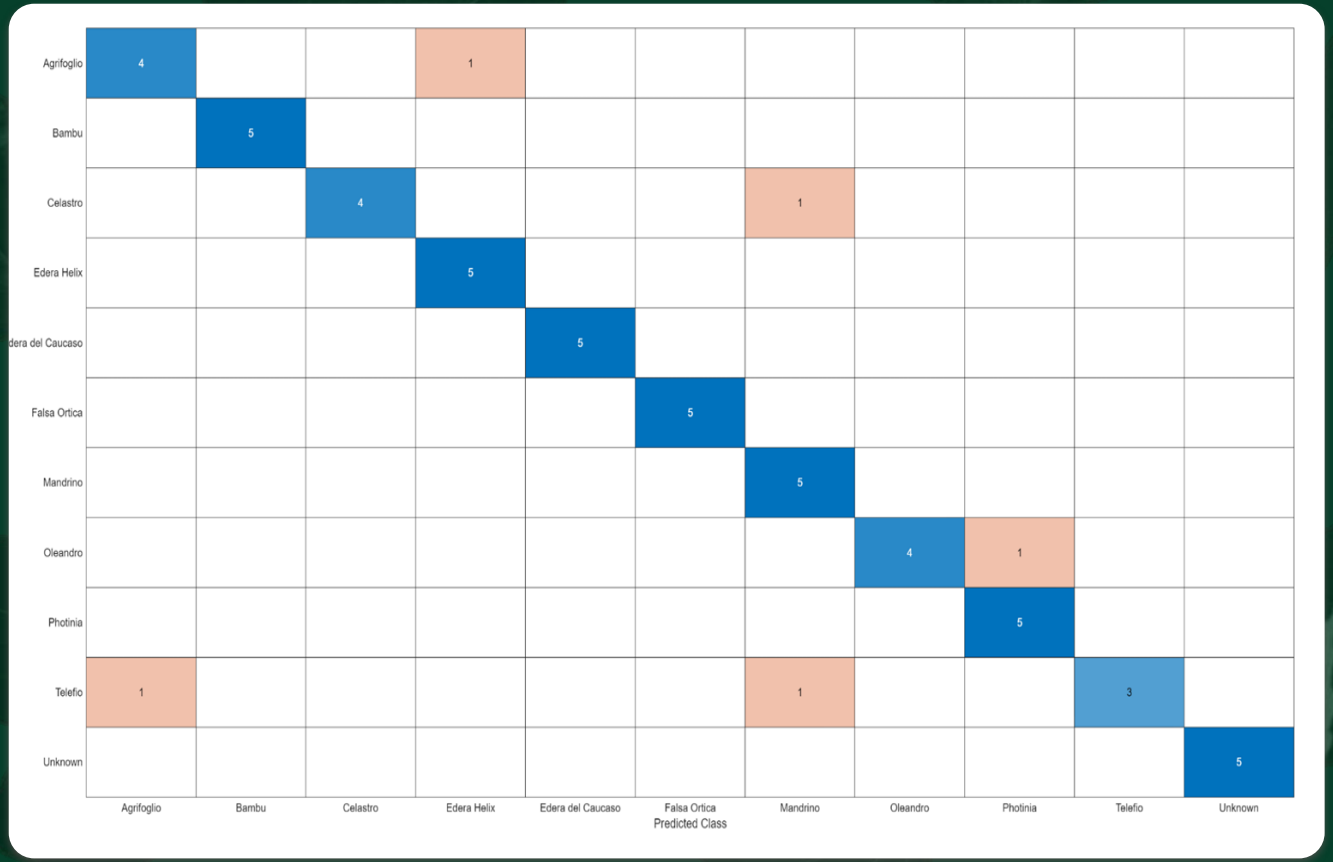
Using Medium Tree

Accuracy	98%
Precision	95%
Recall	99%



On validation set, 5 folds

CFM Model Ensemble



Based on CFM Unknown/Leaf results and CFM Leaves result

Metrics (model ensemble) Using

Weighted-KNN, 10 neighbours, euclidean distance

Accuracy	Precision	Recall
91%	93%	90%

Jaccard Coefficient

Ground truth mask based on polygonal coordinates from COCO json

Jaccard Coefficient

Mask based on Segmentation

Intersection of 2 masks

Union of 2 masks

~ 0.92

(adimensional)

Measure of the similarity between two masks

Problems occured

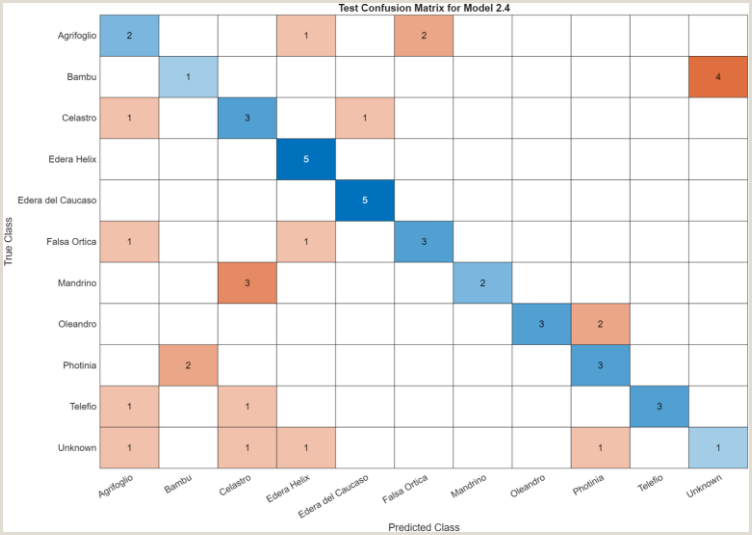
Using almost only **feature-based descriptors,** some leaves with similar shapes are difficult to discriminate based on our features.

Possible solutions: skeleton for the vein structures, LBP for texture

Very similar leaves in shapes

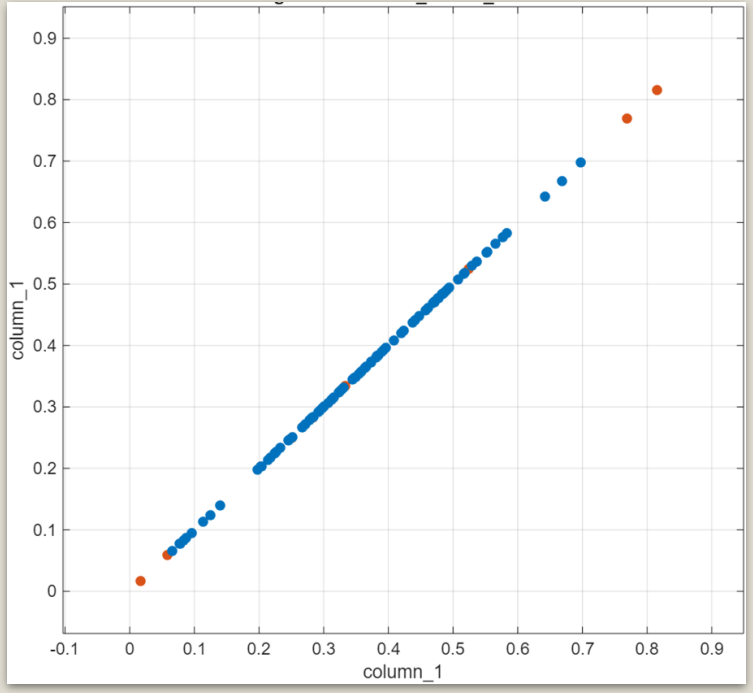
Failed Attempts

Classification of 10 Leaves class and Unknown



10 Leaves and Unknown class model CFM

10 Leaves and Unknown (Model Ensemble) CFM



Scatter plot of train set

Features thresholding on test with train data

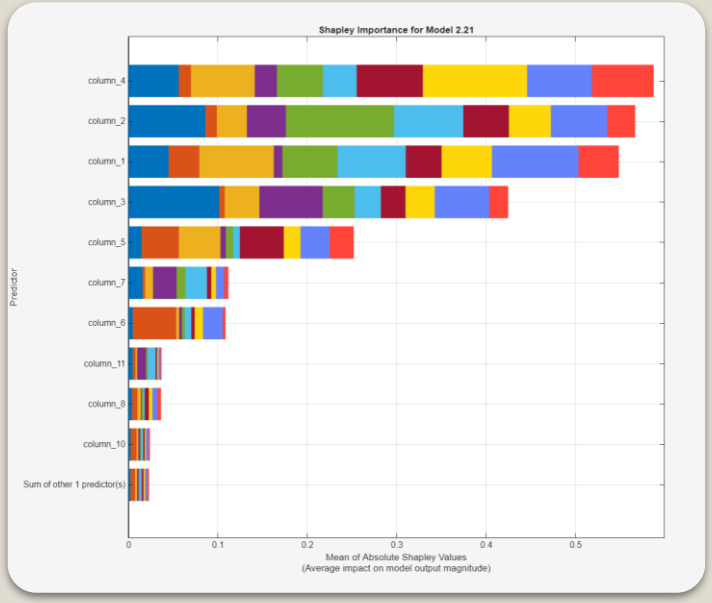
Separate Unknown and Leaf classes using thresholds set with an AND condition, calculated from the training set.

Problem: <u>unknown objects differ significantly from one another</u>, making it hard to determine a universal threshold.

Possibile classification

Using ramification as a leaf feature
Using leaf serrations as a feature
Using all Hu Moments

Problems: the first two were not relevant at all, the third one had risks of overfitting



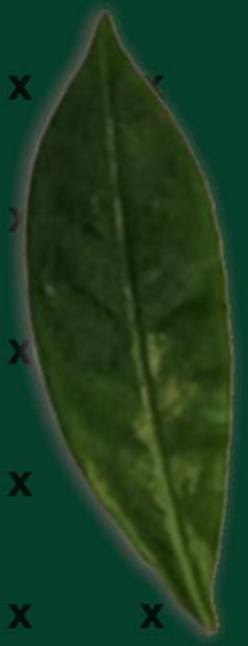
Shapley importance graph containing all 7 Hu Moments

Other features considered

SIS: Shape Irregularity Score. (circularity, mean_distance, curvature)

Skewness_G: indicates whether the image's green channel histogram is unbalanced toward darker or lighter tones.

Kurtosis_G: indicates whether the image's green channel histogram has a high concentration around the mean or is more distributed.



Symmetry Score: Is a score (0-1) for how symmetrical the bbox of the object is with respect to the minor axis

Entropy_value: low entropy indicates uniform areas, high entropy signals images rich in detail and variation

GreenRatio: measures the percentage of pixels in which green prevails over red and blue

Using features such as:

- Symmetry
- Circularity
- Eccentricity
- Solidity
- Perimeter
- Entropy
- Kurtosis
- Skewness
- Curvature
- Green ratio

These features may lead to an overfitted model

Agrifoglio	100.0%									
Bambu		100.0%								
Celastro			100.0%							
Edera Helix				100.0%						
Edera del Caucaso					100.0%					
Falsa Ortica						100.0%				
Mandrino							100.0%			
Oleandro								100.0%		
Photinia									100.0%	
Telefio										100.0%

Overfitting of Test confusion Matrix

Possible upgrades

Further pre-processing on object segmentation is needed for objects with uniform color similar to the background (e.g., improving border distinction).

Refining segmentation for objects with strong reflections and deep shadows.

Resources used

Minimum Bounding Box calculator (link matlab)

Hu Moments calculator (link github)

Feature Extraction and Automatic Recognition of Plant (paper link)